STAT4340-Project1

2026-03-09

Project 1 - Sanjit Subhash

Phase 1

```
library(tidyverse)
```

```
## Warning: package 'purrr' was built under R version 4.5.2
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.2.0     ✔ readr     2.1.5
## ✔ forcats   1.0.1     ✔ stringr   1.6.0
## ✔ ggplot2   4.0.0     ✔ tibble    3.3.0
## ✔ lubridate 1.9.4     ✔ tidyr     1.3.1
## ✔ purrr     1.2.1
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

```
men <- read.csv("/Users/sanjitsubhash/Downloads/trainingData.csv")
women <- read.csv("/Users/sanjitsubhash/Downloads/trainingDataWomen.csv")

dim(men)
```

```
## [1] 1449   40
```

```
dim(women)
```

```
## [1] 961  40
```

```
names(men)
```

```
##  [1] "Season"     "DayNum"     "gameid"     "TeamName_A" "TeamName_B"
##  [6] "TeamID_A"   "TeamID_B"   "Win"        "Score_A"    "Score_B"
## [11] "FGM_A"      "FGM_B"      "FGA_A"      "FGA_B"      "FGM3_A"
## [16] "FGM3_B"     "FGA3_A"     "FGA3_B"     "FTM_A"      "FTM_B"
## [21] "FTA_A"      "FTA_B"      "OR_A"       "OR_B"       "DR_A"
## [26] "DR_B"       "Ast_A"      "Ast_B"      "TO_A"       "TO_B"
## [31] "Stl_A"      "Stl_B"      "Blk_A"      "Blk_B"      "PF_A"
## [36] "PF_B"       "Region_A"   "Seed_A"     "Region_B"   "Seed_B"
```

```
head(men, 3)
```

```
##   Season DayNum gameid    TeamName_A  TeamName_B TeamID_A TeamID_B Win  Score_A
## 1   2003    134      1 UNC Asheville TX Southern     1421     1411   A 71.20690
## 2   2003    136      2       Arizona     Vermont     1112     1436   A 85.21429
## 3   2003    136      3       Memphis  Arizona St     1272     1113   B 74.51724
##    Score_B    FGM_A    FGM_B    FGA_A    FGA_B   FGM3_A   FGM3_B   FGA3_A
## 1 72.80000 24.37931 24.73333 56.79310 55.26667 6.482759 5.933333 18.00000
## 2 67.79310 30.32143 24.82759 65.71429 55.86207 7.035714 5.275862 20.07143
## 3 75.96552 26.27586 27.20690 60.00000 56.89655 7.000000 4.000000 20.06897
##     FGA3_B    FTM_A    FTM_B    FTA_A    FTA_B     OR_A     OR_B     DR_A
## 1 18.50000 15.96552 17.40000 20.93103 28.06667 12.27586 13.16667 23.17241
## 2 15.48276 17.53571 12.86207 25.00000 19.55172 15.17857 12.96552 27.64286
## 3 12.58621 14.96552 17.55172 22.89655 26.20690 14.06897 13.68966 25.96552
##       DR_B    Ast_A    Ast_B     TO_A     TO_B    Stl_A    Stl_B    Blk_A
## 1 24.80000 13.03448 14.20000 16.20690 15.23333 7.068966 6.433333 3.000000
## 2 25.72414 17.64286 14.20690 14.78571 14.06897 8.464286 6.862069 4.214286
## 3 23.31034 16.62069 15.55172 13.79310 14.00000 7.379310 5.206897 5.068966
##      Blk_B     PF_A     PF_B Region_A Seed_A Region_B Seed_B
## 1 2.233333 19.10345 18.30000        X     16        X     16
## 2 2.965517 17.75000 15.89655        Z      1        Z     16
## 3 4.241379 18.75862 19.41379        Z      7        Z     10
```

```
head(women, 3)
```

```
##   Season DayNum gameid TeamName_A TeamName_B TeamID_A TeamID_B Win  Score_A
## 1   2010    138      1     Baylor  Fresno St     3124     3201   A 72.09375
## 2   2010    138      2     Dayton        TCU     3173     3395   A 70.76923
## 3   2010    138      3    Hampton       Duke     3214     3181   B 62.83333
##    Score_B    FGM_A    FGM_B    FGA_A    FGA_B   FGM3_A   FGM3_B   FGA3_A
## 1 73.45455 26.09375 25.84848 56.34375 61.57576 2.468750 8.818182  8.65625
## 2 72.30000 25.88462 25.83333 61.53846 62.26667 5.461538 6.866667 17.46154
## 3 71.59375 22.20000 26.84375 60.10000 63.75000 4.900000 4.562500 16.10000
##     FGA3_B    FTM_A    FTM_B    FTA_A    FTA_B     OR_A     OR_B     DR_A
## 1 24.84848 17.43750 12.93939 24.00000 18.24242 12.18750 13.87879 31.06250
## 2 19.20000 13.53846 13.76667 19.34615 18.40000 14.73077 13.50000 28.07692
## 3 14.46875 13.53333 13.34375 18.93333 20.15625 14.96667 17.71875 22.66667
##       DR_B    Ast_A    Ast_B     TO_A     TO_B     Stl_A     Stl_B    Blk_A
## 1 24.57576 14.68750 14.63636 16.87500 15.63636  6.968750 10.393939 7.250000
## 2 25.83333 14.46154 15.60000 16.96154 16.03333  7.730769  9.833333 4.076923
## 3 25.59375 11.60000 14.25000 15.53333 17.90625 10.100000 13.375000 2.233333
##      Blk_B     PF_A     PF_B Region_A Seed_A Region_B Seed_B
## 1 2.606061 14.12500 16.39394        X      4        X     13
## 2 3.600000 16.57692 16.23333        X      8        X      9
## 3 5.281250 18.83333 17.68750        X     15        X      2
```

```
table(men$Win)
```

```
##
##   A   B
## 727 722
```

```
table(women$Win)
```

```
##
##   A   B
## 486 475
```

The men’s dataset had 1449 observations and the women’s dataset had 961 observations. I also checked the response variable Win and found that the classes were very balanced in both datasets, which is helpful for classification modeling.

Phase 2:

```
men2 <- men %>%
  mutate(
    y = ifelse(Win == "A", 1, 0),
    SeedDiff = Seed_B - Seed_A,
    ScoreDiff = Score_A - Score_B,
    FGMDiff = FGM_A - FGM_B,
    FGADiff = FGA_A - FGA_B,
    FGM3Diff = FGM3_A - FGM3_B,
    FGA3Diff = FGA3_A - FGA3_B,
    FTMDiff = FTM_A - FTM_B,
    FTADiff = FTA_A - FTA_B,
    ORDiff = OR_A - OR_B,
    DRDiff = DR_A - DR_B,
    AstDiff = Ast_A - Ast_B,
    TODiff = TO_A - TO_B,
    StlDiff = Stl_A - Stl_B,
    BlkDiff = Blk_A - Blk_B,
    PFDiff = PF_A - PF_B
  ) %>%
  select(
    y, SeedDiff, ScoreDiff, FGMDiff, FGADiff, FGM3Diff, FGA3Diff,
    FTMDiff, FTADiff, ORDiff, DRDiff, AstDiff, TODiff,
    StlDiff, BlkDiff, PFDiff
  )

women2 <- women %>%
  mutate(
    y = ifelse(Win == "A", 1, 0),
    SeedDiff = Seed_B - Seed_A,
    ScoreDiff = Score_A - Score_B,
    FGMDiff = FGM_A - FGM_B,
    FGADiff = FGA_A - FGA_B,
    FGM3Diff = FGM3_A - FGM3_B,
    FGA3Diff = FGA3_A - FGA3_B,
    FTMDiff = FTM_A - FTM_B,
    FTADiff = FTA_A - FTA_B,
    ORDiff = OR_A - OR_B,
    DRDiff = DR_A - DR_B,
    AstDiff = Ast_A - Ast_B,
    TODiff = TO_A - TO_B,
    StlDiff = Stl_A - Stl_B,
    BlkDiff = Blk_A - Blk_B,
    PFDiff = PF_A - PF_B
  ) %>%
  select(
    y, SeedDiff, ScoreDiff, FGMDiff, FGADiff, FGM3Diff, FGA3Diff,
    FTMDiff, FTADiff, ORDiff, DRDiff, AstDiff, TODiff,
    StlDiff, BlkDiff, PFDiff
  )
```

```
head(men2, 3)
```

```
##   y SeedDiff ScoreDiff    FGMDiff  FGADiff  FGM3Diff  FGA3Diff   FTMDiff
## 1 1        0 -1.593103 -0.3540230 1.526437 0.5494253 -0.500000 -1.434483
## 2 1       15 17.421182  5.4938424 9.852217 1.7598522  4.588670  4.673645
## 3 0        3 -1.448276 -0.9310345 3.103448 3.0000000  7.482759 -2.586207
##     FTADiff     ORDiff    DRDiff   AstDiff     TODiff   StlDiff   BlkDiff
## 1 -7.135632 -0.8908046 -1.627586 -1.165517  0.9735632 0.6356322 0.7666667
## 2  5.448276  2.2130542  1.918719  3.435961  0.7167488 1.6022167 1.2487685
## 3 -3.310345  0.3793103  2.655172  1.068966 -0.2068966 2.1724138 0.8275862
##       PFDiff
## 1  0.8034483
## 2  1.8534483
## 3 -0.6551724
```

```
head(women2, 3)
```

```
##   y SeedDiff ScoreDiff     FGMDiff    FGADiff  FGM3Diff   FGA3Diff    FTMDiff
## 1 1        9 -1.360795  0.24526515 -5.2320076 -6.349432 -16.192235  4.4981061
## 2 1        1 -1.530769  0.05128205 -0.7282051 -1.405128  -1.738462 -0.2282051
## 3 0      -13 -8.760417 -4.64375000 -3.6500000  0.337500   1.631250  0.1895833
##      FTADiff    ORDiff    DRDiff     AstDiff     TODiff   StlDiff    BlkDiff
## 1  5.7575758 -1.691288  6.486742  0.05113636  1.2386364 -3.425189  4.6439394
## 2  0.9461538  1.230769  2.243590 -1.13846154  0.9282051 -2.102564  0.4769231
## 3 -1.2229167 -2.752083 -2.927083 -2.65000000 -2.3729167 -3.275000 -3.0479167
##       PFDiff
## 1 -2.2689394
## 2  0.3435897
## 3  1.1458333
```

```
summary(men2)
```

```
##        y             SeedDiff           ScoreDiff            FGMDiff
##  Min.   :0.0000   Min.   :-15.00000   Min.   :-22.89286   Min.   :-9.357143
##  1st Qu.:0.0000   1st Qu.: -5.00000   1st Qu.: -4.86720   1st Qu.:-1.992943
##  Median :1.0000   Median :  0.00000   Median :  0.12903   Median :-0.057576
##  Mean   :0.5017   Mean   :  0.05383   Mean   :  0.02173   Mean   : 0.004749
##  3rd Qu.:1.0000   3rd Qu.:  6.00000   3rd Qu.:  4.71385   3rd Qu.: 1.965726
##  Max.   :1.0000   Max.   : 15.00000   Max.   : 21.17380   Max.   : 8.677656
##     FGADiff             FGM3Diff            FGA3Diff
##  Min.   :-22.07143   Min.   :-5.246323   Min.   :-12.42789
##  1st Qu.: -3.14576   1st Qu.:-1.220919   1st Qu.: -2.80370
##  Median :  0.03030   Median : 0.031250   Median :  0.11029
##  Mean   : -0.07655   Mean   : 0.004629   Mean   : -0.02359
##  3rd Qu.:  3.02763   3rd Qu.: 1.201299   3rd Qu.:  2.75000
##  Max.   : 14.03676   Max.   : 6.062500   Max.   : 14.27849
##     FTMDiff              FTADiff              ORDiff             DRDiff
##  Min.   :-10.205882   Min.   :-13.09412   Min.   :-9.21429   Min.   :-9.28571
##  1st Qu.: -1.819853   1st Qu.: -2.26916   1st Qu.:-1.62745   1st Qu.:-1.60478
##  Median :  0.011029   Median : -0.02723   Median : 0.03125   Median : 0.06262
##  Mean   :  0.007607   Mean   :  0.02712   Mean   : 0.02701   Mean   : 0.08905
##  3rd Qu.:  1.857143   3rd Qu.:  2.49496   3rd Qu.: 1.72371   3rd Qu.: 1.76723
##  Max.   :  8.059629   Max.   : 10.82759   Max.   : 7.69223   Max.   : 9.01078
##     AstDiff             TODiff            StlDiff            BlkDiff
##  Min.   :-9.23265   Min.   :-7.58065   Min.   :-5.70370   Min.   :-6.24242
##  1st Qu.:-1.71836   1st Qu.:-1.23529   1st Qu.:-1.34375   1st Qu.:-1.12745
##  Median : 0.04963   Median : 0.06345   Median :-0.07116   Median : 0.02941
##  Mean   : 0.04597   Mean   : 0.10452   Mean   :-0.05219   Mean   : 0.02680
##  3rd Qu.: 1.86364   3rd Qu.: 1.44497   3rd Qu.: 1.19556   3rd Qu.: 1.10511
##  Max.   : 9.22014   Max.   : 7.66667   Max.   : 6.28368   Max.   : 6.75269
##      PFDiff
##  Min.   :-7.66488
##  1st Qu.:-1.73939
##  Median : 0.02941
##  Mean   : 0.02083
##  3rd Qu.: 1.73529
##  Max.   : 7.89655
```

```
summary(women2)
```

```
##        y             SeedDiff           ScoreDiff           FGMDiff
##  Min.   :0.0000   Min.   :-15.00000   Min.   :-30.8485   Min.   :-13.84848
##  1st Qu.:0.0000   1st Qu.: -5.00000   1st Qu.: -6.1251   1st Qu.: -2.38310
##  Median :1.0000   Median :  0.00000   Median :  0.6172   Median :  0.05128
##  Mean   :0.5057   Mean   : -0.02289   Mean   :  0.1315   Mean   :  0.03579
##  3rd Qu.:1.0000   3rd Qu.:  5.00000   3rd Qu.:  6.3333   3rd Qu.:  2.65806
##  Max.   :1.0000   Max.   : 15.00000   Max.   : 27.0235   Max.   : 12.08016
##     FGADiff             FGM3Diff          FGA3Diff           FTMDiff
##  Min.   :-14.63352   Min.   :-7.6061   Min.   :-22.3448   Min.   :-8.58807
##  1st Qu.: -3.49539   1st Qu.:-1.3812   1st Qu.: -3.5991   1st Qu.:-2.03629
##  Median : -0.15152   Median : 0.1241   Median :  0.3317   Median : 0.00000
##  Mean   :  0.01606   Mean   : 0.1084   Mean   :  0.2670   Mean   :-0.04855
##  3rd Qu.:  3.45636   3rd Qu.: 1.6129   3rd Qu.:  4.2333   3rd Qu.: 1.82964
##  Max.   : 17.60000   Max.   : 7.0606   Max.   : 19.9592   Max.   : 8.10968
##     FTADiff              ORDiff             DRDiff
##  Min.   :-12.00473   Min.   :-8.56250   Min.   :-11.595833
##  1st Qu.: -2.71075   1st Qu.:-1.99091   1st Qu.: -2.329255
##  Median : -0.03125   Median : 0.05767   Median :  0.008065
##  Mean   : -0.04199   Mean   : 0.05701   Mean   :  0.003471
##  3rd Qu.:  2.50379   3rd Qu.: 2.09677   3rd Qu.:  2.321691
##  Max.   : 11.09892   Max.   : 9.28476   Max.   : 10.250000
##     AstDiff              TODiff            StlDiff            BlkDiff
##  Min.   :-10.12853   Min.   :-7.9044   Min.   :-8.59892   Min.   :-6.90909
##  1st Qu.: -2.03664   1st Qu.:-1.5161   1st Qu.:-1.88609   1st Qu.:-1.15017
##  Median :  0.09879   Median : 0.1396   Median : 0.01287   Median : 0.04738
##  Mean   :  0.15347   Mean   : 0.1203   Mean   :-0.03608   Mean   : 0.04364
##  3rd Qu.:  2.50417   3rd Qu.: 1.7977   3rd Qu.: 1.74879   3rd Qu.: 1.29032
##  Max.   : 11.55882   Max.   : 8.4750   Max.   : 7.25181   Max.   : 5.68750
##      PFDiff
##  Min.   :-9.463947
##  1st Qu.:-1.796371
##  Median : 0.004735
##  Mean   : 0.043031
##  3rd Qu.: 1.903226
##  Max.   : 9.233871
```

I felt this was a cleaner way to represent the matchup, since the model then focuses on how Team A compares to Team B rather than treating the raw team labels separately.

Phase 3

```
table(men2$y)
```

```
##
##   0   1
## 722 727
```

```
table(women2$y)
```

```
##
##   0   1
## 475 486
```

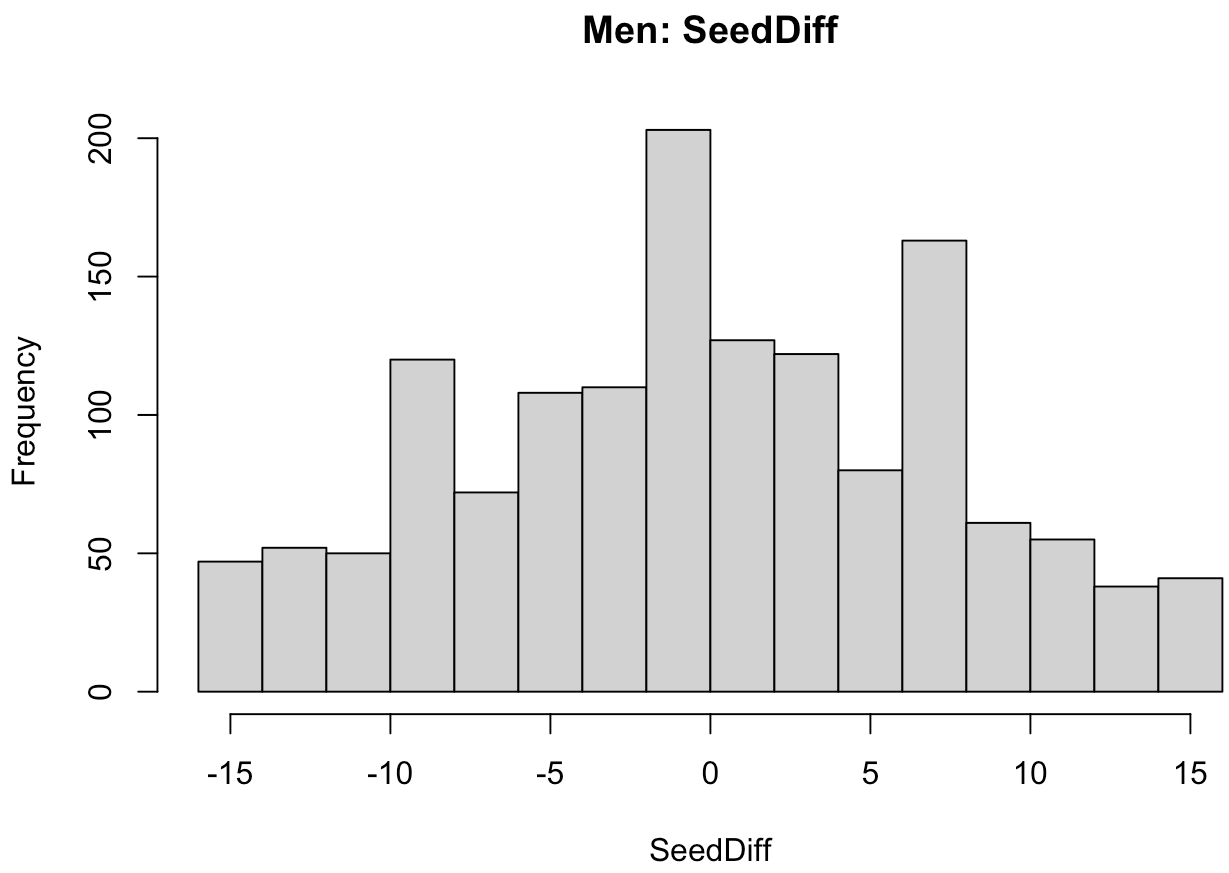
```
summary(men2[, c("SeedDiff", "ScoreDiff", "AstDiff", "TODiff", "StlDiff")])
```

```
##     SeedDiff           ScoreDiff            AstDiff             TODiff
##  Min.   :-15.00000   Min.   :-22.89286   Min.   :-9.23265   Min.   :-7.58065
##  1st Qu.: -5.00000   1st Qu.: -4.86720   1st Qu.:-1.71836   1st Qu.:-1.23529
##  Median :  0.00000   Median :  0.12903   Median : 0.04963   Median : 0.06345
##  Mean   :  0.05383   Mean   :  0.02173   Mean   : 0.04597   Mean   : 0.10452
##  3rd Qu.:  6.00000   3rd Qu.:  4.71385   3rd Qu.: 1.86364   3rd Qu.: 1.44497
##  Max.   : 15.00000   Max.   : 21.17380   Max.   : 9.22014   Max.   : 7.66667
##     StlDiff
##  Min.   :-5.70370
##  1st Qu.:-1.34375
##  Median :-0.07116
##  Mean   :-0.05219
##  3rd Qu.: 1.19556
##  Max.   : 6.28368
```

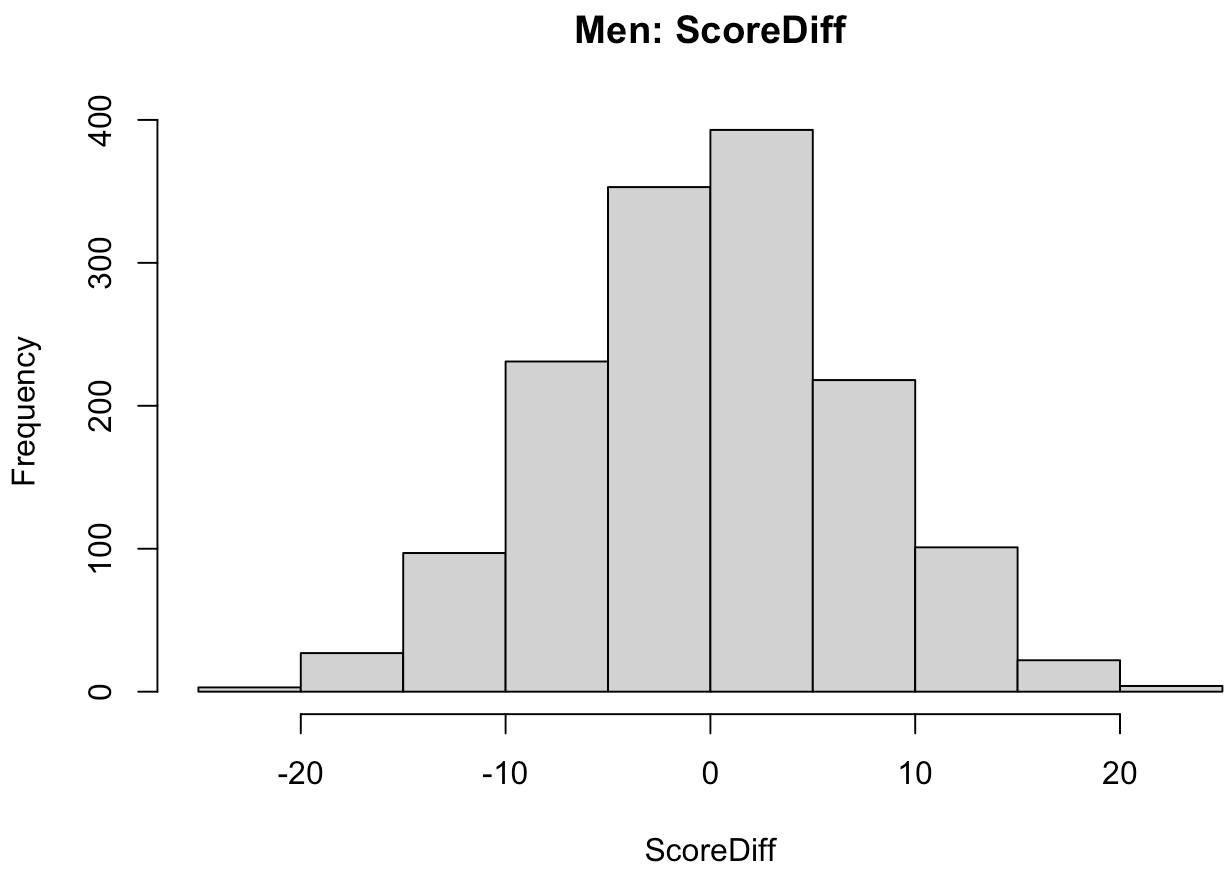
```
summary(women2[, c("SeedDiff", "ScoreDiff", "AstDiff", "TODiff", "StlDiff")])
```

```
##     SeedDiff           ScoreDiff           AstDiff              TODiff
##  Min.   :-15.00000   Min.   :-30.8485   Min.   :-10.12853   Min.   :-7.9044
##  1st Qu.: -5.00000   1st Qu.: -6.1251   1st Qu.: -2.03664   1st Qu.:-1.5161
##  Median :  0.00000   Median :  0.6172   Median :  0.09879   Median : 0.1396
##  Mean   : -0.02289   Mean   :  0.1315   Mean   :  0.15347   Mean   : 0.1203
##  3rd Qu.:  5.00000   3rd Qu.:  6.3333   3rd Qu.:  2.50417   3rd Qu.: 1.7977
##  Max.   : 15.00000   Max.   : 27.0235   Max.   : 11.55882   Max.   : 8.4750
##     StlDiff
##  Min.   :-8.59892
##  1st Qu.:-1.88609
##  Median : 0.01287
##  Mean   :-0.03608
##  3rd Qu.: 1.74879
##  Max.   : 7.25181
```

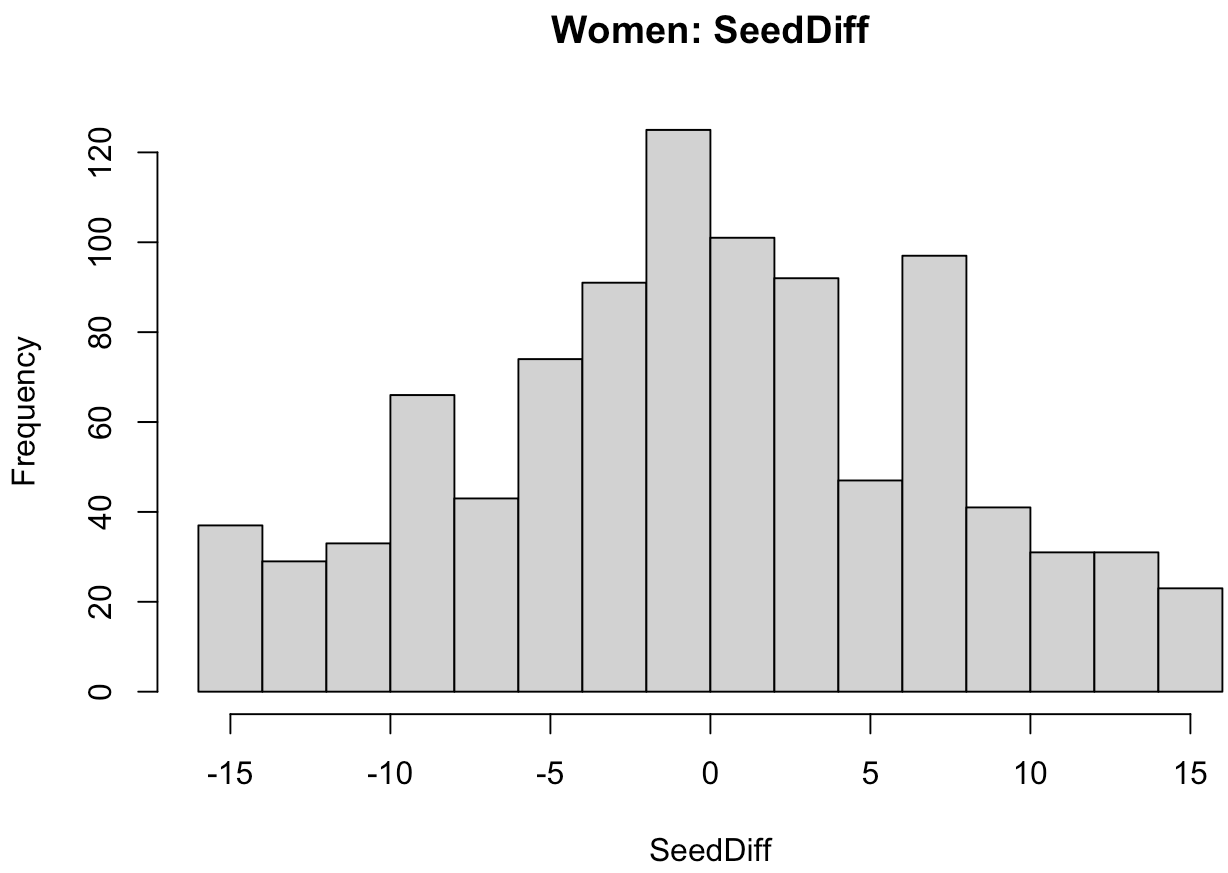
```
hist(men2$SeedDiff, main = "Men: SeedDiff", xlab = "SeedDiff")
```



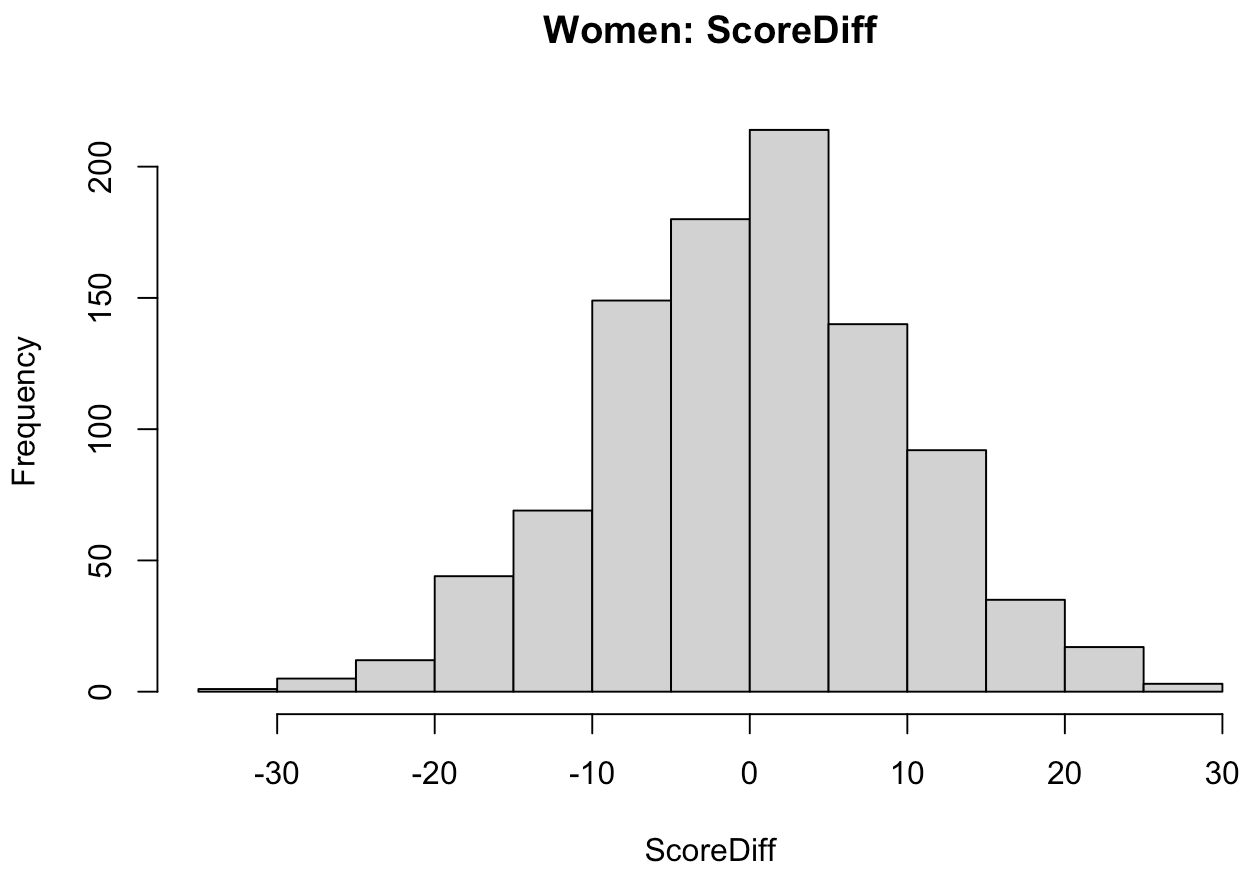

```
hist(men2$ScoreDiff, main = "Men: ScoreDiff", xlab = "ScoreDiff")
```




```
hist(women2$SeedDiff, main = "Women: SeedDiff", xlab = "SeedDiff")
```

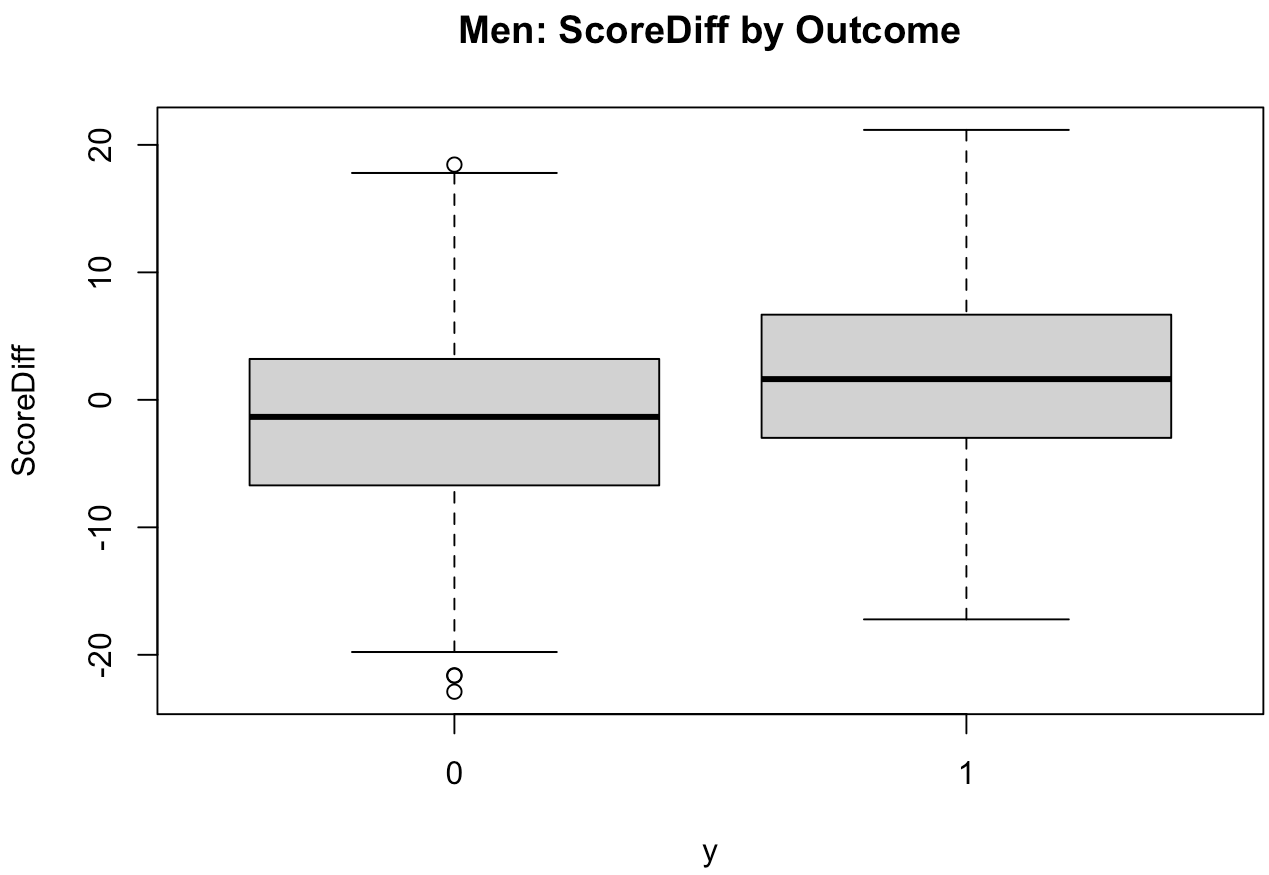



```
hist(women2$ScoreDiff, main = "Women: ScoreDiff", xlab = "ScoreDiff")
```

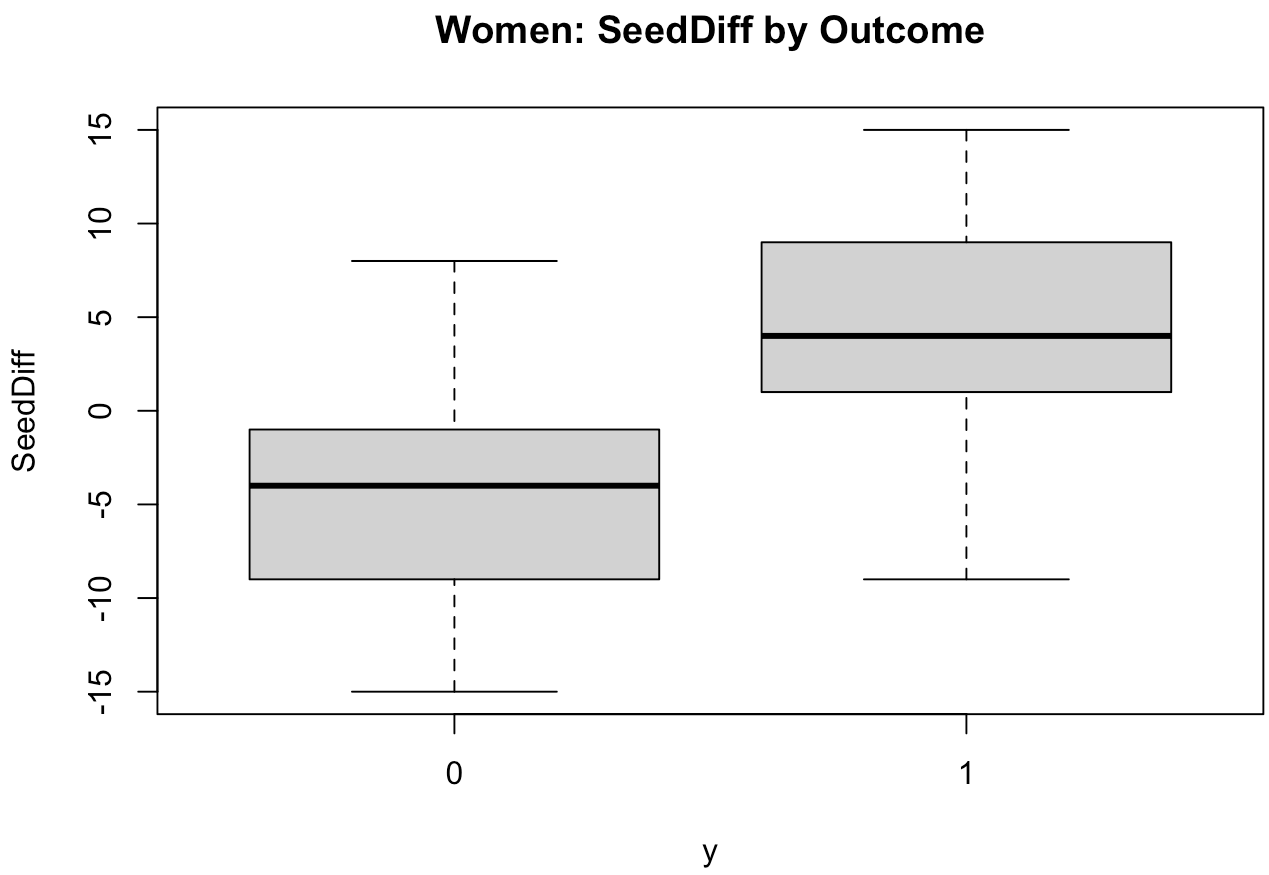



```
boxplot(SeedDiff ~ y, data = men2, main = "Men: SeedDiff by Outcome")
```

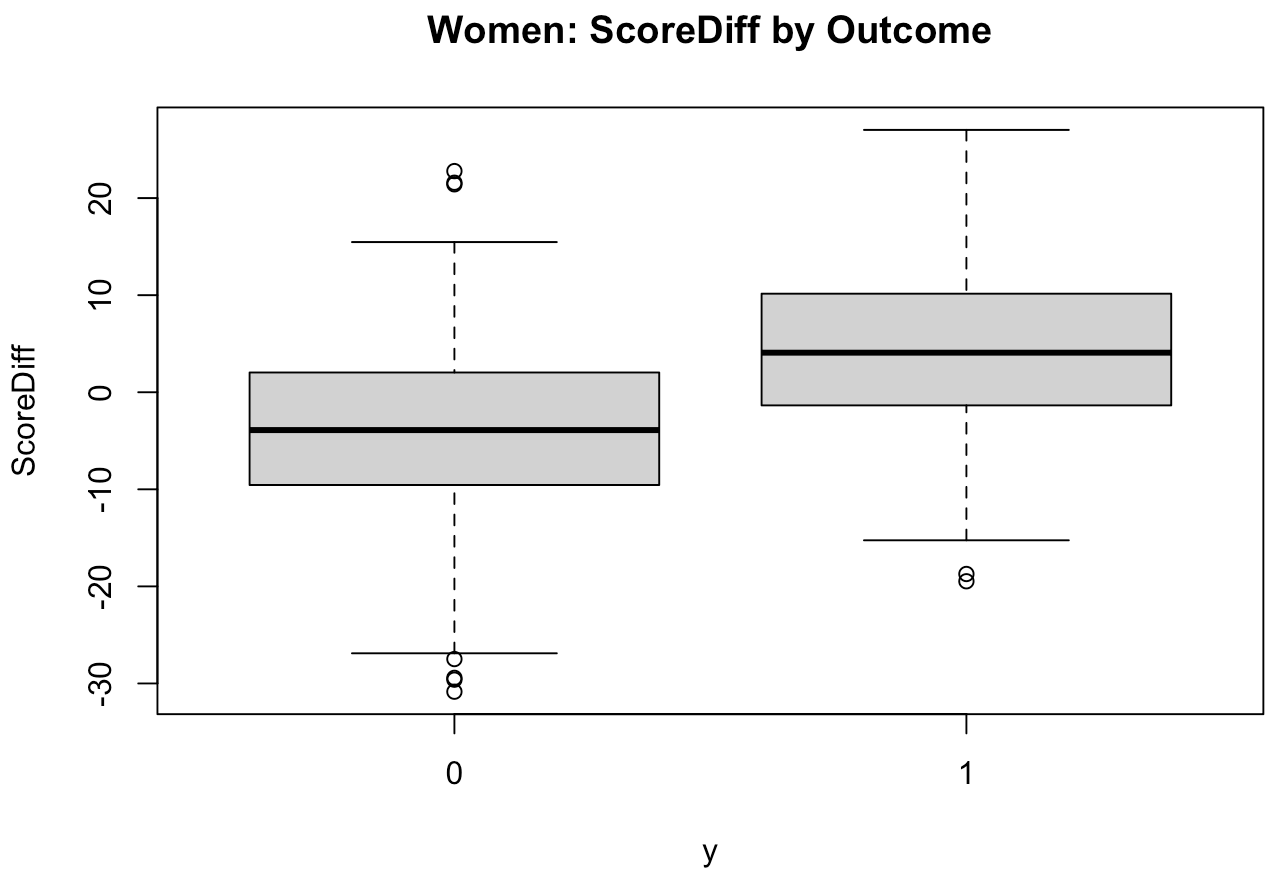
```
boxplot(ScoreDiff ~ y, data = men2, main = "Men: ScoreDiff by Outcome")
```




```
boxplot(SeedDiff ~ y, data = women2, main = "Women: SeedDiff by Outcome")
```




```
boxplot(ScoreDiff ~ y, data = women2, main = "Women: ScoreDiff by Outcome")
```




```
cor(men2[, -1])
```

```
##              SeedDiff  ScoreDiff     FGMDiff    FGADiff     FGM3Diff
## SeedDiff   1.00000000 0.42560543  0.43914683 0.29727890  0.112259736
## ScoreDiff  0.42560543 1.00000000  0.91730002 0.73163144  0.380215950
## FGMDiff    0.43914683 0.91730002  1.00000000 0.76801687  0.228931253
## FGADiff    0.29727890 0.73163144  0.76801687 1.00000000  0.204029350
## FGM3Diff   0.11225974 0.38021595  0.22893125 0.20402935  1.000000000
## FGA3Diff   0.04456056 0.26856861  0.11490714 0.25974601  0.916722832
## FTMDiff    0.13235474 0.47892610  0.17904117 0.19196493 -0.133669691
## FTADiff    0.10521787 0.42393633  0.17747625 0.24243293 -0.231765157
## ORDiff     0.27916215 0.33400164  0.34371858 0.65079615 -0.196771684
## DRDiff     0.31158070 0.49321373  0.49119219 0.35798248  0.016387869
## AstDiff    0.41093777 0.62642824  0.70032400 0.43100000  0.269979789
## TODiff    -0.22002531 0.07936108  0.03422588 0.04355232 -0.244160019
## StlDiff    0.09638991 0.25469173  0.27433495 0.42332314 -0.007417536
## BlkDiff    0.34693647 0.22983875  0.28800617 0.31699356 -0.167659552
## PFDiff    -0.22458132 0.08836000 -0.02393088 0.18702213 -0.045592883
##              FGA3Diff      FTMDiff     FTADiff     ORDiff      DRDiff
## SeedDiff   0.04456056  0.132354736  0.10521787  0.2791622  0.31158070
## ScoreDiff  0.26856861  0.478926103  0.42393633  0.3340016  0.49321373
## FGMDiff    0.11490714  0.179041166  0.17747625  0.3437186  0.49119219
## FGADiff    0.25974601  0.191964930  0.24243293  0.6507962  0.35798248
## FGM3Diff   0.91672283 -0.133669691 -0.23176516 -0.1967717  0.01638787
## FGA3Diff   1.00000000 -0.134847515 -0.19633315 -0.0878255 -0.03846454
## FTMDiff   -0.13484751  1.000000000  0.92039814  0.2966953  0.26761405
## FTADiff   -0.19633315  0.920398143  1.00000000  0.4453383  0.28799777
## ORDiff    -0.08782550  0.296695326  0.44533826  1.0000000  0.21460460
## DRDiff    -0.03846454  0.267614053  0.28799777  0.2146046  1.00000000
## AstDiff    0.15567996  0.008708073 -0.01476457  0.1611809  0.35585652
## TODiff    -0.20904225  0.304671846  0.40918720  0.2859576  0.20069499
## StlDiff    0.07121201  0.104399772  0.18361788  0.3087771 -0.21521742
## BlkDiff   -0.15381972  0.115201431  0.20205563  0.3870866  0.39489789
## PFDiff     0.02938549  0.320873394  0.38010767  0.2608214 -0.15394891
##                AstDiff      TODiff      StlDiff     BlkDiff      PFDiff
## SeedDiff   0.410937768 -0.22002531  0.096389912  0.34693647 -0.22458132
## ScoreDiff  0.626428235  0.07936108  0.254691725  0.22983875  0.08836000
## FGMDiff    0.700324004  0.03422588  0.274334953  0.28800617 -0.02393088
## FGADiff    0.431000004  0.04355232  0.423323138  0.31699356  0.18702213
## FGM3Diff   0.269979789 -0.24416002 -0.007417536 -0.16765955 -0.04559288
## FGA3Diff   0.155679955 -0.20904225  0.071212013 -0.15381972  0.02938549
## FTMDiff    0.008708073  0.30467185  0.104399772  0.11520143  0.32087339
## FTADiff   -0.014764575  0.40918720  0.183617884  0.20205563  0.38010767
## ORDiff     0.161180881  0.28595756  0.308777075  0.38708661  0.26082137
## DRDiff     0.355856522  0.20069499 -0.215217419  0.39489789 -0.15394891
## AstDiff    1.000000000  0.03675886  0.144302454  0.14897278 -0.10319356
## TODiff     0.036758857  1.00000000  0.213351917  0.15445062  0.41902034
## StlDiff    0.144302454  0.21335192  1.000000000  0.19906688  0.31388315
## BlkDiff    0.148972783  0.15445062  0.199066882  1.00000000 -0.05161742
## PFDiff    -0.103193563  0.41902034  0.313883152 -0.05161742  1.00000000
```

```
cor(women2[, -1])
```

```
##              SeedDiff  ScoreDiff     FGMDiff     FGADiff    FGM3Diff
## SeedDiff   1.00000000  0.5989420  0.63112651  0.38129145 -0.07476998
## ScoreDiff  0.59894204  1.0000000  0.94803074  0.68604065  0.27885523
## FGMDiff    0.63112651  0.9480307  1.00000000  0.69877794  0.12691671
## FGADiff    0.38129145  0.6860406  0.69877794  1.00000000  0.10280711
## FGM3Diff  -0.07476998  0.2788552  0.12691671  0.10280711  1.00000000
## FGA3Diff  -0.20491970  0.1166556 -0.03934086  0.11652981  0.94102001
## FTMDiff    0.27727367  0.4311868  0.23237828  0.23041849 -0.25080621
## FTADiff    0.24392354  0.3151909  0.15726713  0.24882502 -0.37365894
## ORDiff     0.26344439  0.2361189  0.28125862  0.63597032 -0.34824048
## DRDiff     0.38874372  0.4679169  0.45589636  0.23654102  0.06248390
## AstDiff    0.43466803  0.7258824  0.77373897  0.37741393  0.25530726
## TODiff    -0.16514696 -0.2055118 -0.21620394 -0.06267482 -0.27580503
## StlDiff    0.05262131  0.2319789  0.26699423  0.41350187 -0.09767158
## BlkDiff    0.42639037  0.3119320  0.35633215  0.23490314 -0.14780580
## PFDiff    -0.24528098 -0.2134291 -0.26470760  0.16330475 -0.16404981
##              FGA3Diff     FTMDiff     FTADiff      ORDiff      DRDiff
## SeedDiff  -0.20491970  0.27727367  0.24392354  0.26344439  0.38874372
## ScoreDiff  0.11665558  0.43118677  0.31519094  0.23611895  0.46791692
## FGMDiff   -0.03934086  0.23237828  0.15726713  0.28125862  0.45589636
## FGADiff    0.11652981  0.23041849  0.24882502  0.63597032  0.23654102
## FGM3Diff   0.94102001 -0.25080621 -0.37365894 -0.34824048  0.06248390
## FGA3Diff   1.00000000 -0.27350409 -0.35735400 -0.27672008 -0.03398970
## FTMDiff   -0.27350409  1.00000000  0.92508896  0.28075477  0.22291155
## FTADiff   -0.35735400  0.92508896  1.00000000  0.43293021  0.17850201
## ORDiff    -0.27672008  0.28075477  0.43293021  1.00000000  0.07481850
## DRDiff    -0.03398970  0.22291155  0.17850201  0.07481850  1.00000000
## AstDiff    0.10409463  0.02354565 -0.08674876 -0.01483116  0.42123608
## TODiff    -0.20324644  0.15404919  0.25245270  0.24836810 -0.06943848
## StlDiff   -0.05757427  0.09971892  0.17024315  0.32842517 -0.39615611
## BlkDiff   -0.20203033  0.15545729  0.18821766  0.25800039  0.49612152
## PFDiff    -0.04677965  0.17368377  0.27496616  0.29627120 -0.40658862
##               AstDiff      TODiff       StlDiff       BlkDiff      PFDiff
## SeedDiff   0.43466803 -0.16514696  0.0526213061  0.4263903689 -0.24528098
## ScoreDiff  0.72588237 -0.20551176  0.2319788624  0.3119319756 -0.21342912
## FGMDiff    0.77373897 -0.21620394  0.2669942253  0.3563321540 -0.26470760
## FGADiff    0.37741393 -0.06267482  0.4135018662  0.2349031407  0.16330475
## FGM3Diff   0.25530726 -0.27580503 -0.0976715772 -0.1478058020 -0.16404981
## FGA3Diff   0.10409463 -0.20324644 -0.0575742671 -0.2020303301 -0.04677965
## FTMDiff    0.02354565  0.15404919  0.0997189226  0.1554572879  0.17368377
## FTADiff   -0.08674876  0.25245270  0.1702431510  0.1882176594  0.27496616
## ORDiff    -0.01483116  0.24836810  0.3284251681  0.2580003884  0.29627120
## DRDiff     0.42123608 -0.06943848 -0.3961561125  0.4961215223 -0.40658862
## AstDiff    1.00000000 -0.19099172  0.1615478171  0.2544283390 -0.35628875
## TODiff    -0.19099172  1.00000000  0.1985047005 -0.0570477595  0.50261676
## StlDiff    0.16154782  0.19850470  1.0000000000  0.0006167497  0.28592597
## BlkDiff    0.25442834 -0.05704776  0.0006167497  1.0000000000 -0.22827456
## PFDiff    -0.35628875  0.50261676  0.2859259745 -0.2282745647  1.00000000
```

Phase 4: First baseline models

(I) create factor response for caret/tree work

```
men2$y_factor <- factor(ifelse(men2$y == 1, "Awin", "Bwin"))
women2$y_factor <- factor(ifelse(women2$y == 1, "Awin", "Bwin"))

table(men2$y_factor)
```

```
##
## Awin Bwin
##  727  722
```

```
table(women2$y_factor)
```

```
##
## Awin Bwin
##  486  475
```

(II) Fit the first logistic regression

```
glm_men_small <- glm(
  y ~ SeedDiff + ScoreDiff + AstDiff + TODiff + StlDiff,
  data = men2,
  family = binomial
)

summary(glm_men_small)
```

```
##
## Call:
## glm(formula = y ~ SeedDiff + ScoreDiff + AstDiff + TODiff + StlDiff,
##     family = binomial, data = men2)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.0079566  0.0605172   0.131   0.8954
## SeedDiff     0.1501904  0.0106684  14.078   <2e-16 ***
## ScoreDiff    0.0134702  0.0112867   1.193   0.2327
## AstDiff      0.0007375  0.0300159   0.025   0.9804
## TODiff      -0.1001980  0.0319319  -3.138   0.0017 **
## StlDiff      0.0723238  0.0337262   2.144   0.0320 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2008.7  on 1448  degrees of freedom
## Residual deviance: 1620.9  on 1443  degrees of freedom
## AIC: 1632.9
##
## Number of Fisher Scoring iterations: 4
```

(III) Get predicted probabilities and classifications

```
men_probs_small <- predict(glm_men_small, type = "response")
men_pred_small <- ifelse(men_probs_small > 0.5, 1, 0)

table(Predicted = men_pred_small, Actual = men2$y)
```

```
##          Actual
## Predicted   0   1
##         0 505 212
##         1 217 515
```

```
mean(men_pred_small == men2$y)
```

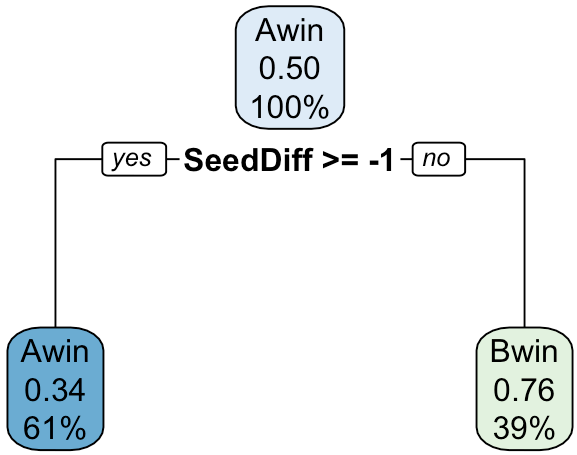
```
## [1] 0.7039337
```

(IV) Fit the first classification tree

```
library(rpart)
library(rpart.plot)
```

```
## Warning: package 'rpart.plot' was built under R version 4.5.2
```

```
tree_men <- rpart(
  y_factor ~ SeedDiff + ScoreDiff + AstDiff + TODiff + StlDiff,
  data = men2,
  method = "class"
)

rpart.plot(tree_men)
```




```
printcp(tree_men)
```

```
##
## Classification tree:
## rpart(formula = y_factor ~ SeedDiff + ScoreDiff + AstDiff + TODiff +
##     StlDiff, data = men2, method = "class")
##
## Variables actually used in tree construction:
## [1] SeedDiff
##
## Root node error: 722/1449 = 0.49827
##
## n= 1449
##
##        CP nsplit rel error  xerror     xstd
## 1 0.39751      0   1.00000 1.06233 0.026316
## 2 0.01000      1   0.60249 0.63158 0.024484
```

```
tree_pred_class <- predict(tree_men, type = "class")
table(Predicted = tree_pred_class, Actual = men2$y_factor)
```

```
##          Actual
## Predicted Awin Bwin
##      Awin  591  299
##      Bwin  136  423
```

```
mean(tree_pred_class == men2$y_factor)
```

```
## [1] 0.699793
```

Phase 5: Expanded Model Comparison

Men full logistic regression

```
glm_men_full <- glm(
  y ~ SeedDiff + ScoreDiff + FGMDiff + FGADiff + FGM3Diff + FGA3Diff +
    FTMDiff + FTADiff + ORDiff + DRDiff + AstDiff + TODiff +
    StlDiff + BlkDiff + PFDiff,
  data = men2,
  family = binomial
)

summary(glm_men_full)
```

```
##
## Call:
## glm(formula = y ~ SeedDiff + ScoreDiff + FGMDiff + FGADiff +
##     FGM3Diff + FGA3Diff + FTMDiff + FTADiff + ORDiff + DRDiff +
##     AstDiff + TODiff + StlDiff + BlkDiff + PFDiff, family = binomial,
##     data = men2)
##
## Coefficients: (1 not defined because of singularities)
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.0005605  0.0612820   0.009 0.992703
## SeedDiff     0.1323680  0.0118844  11.138  < 2e-16 ***
## ScoreDiff    0.1381998  0.0650652   2.124 0.033668 *
## FGMDiff     -0.0524709  0.1386120  -0.379 0.705026
## FGADiff     -0.1345730  0.0391675  -3.436 0.000591 ***
## FGM3Diff    -0.1667246  0.1329176  -1.254 0.209717
## FGA3Diff     0.0304843  0.0463018   0.658 0.510292
## FTMDiff             NA         NA      NA       NA
## FTADiff     -0.1505474  0.0539337  -2.791 0.005249 **
## ORDiff       0.1832167  0.0486079   3.769 0.000164 ***
## DRDiff      -0.0043263  0.0379565  -0.114 0.909253
## AstDiff     -0.0622152  0.0356293  -1.746 0.080780 .
## TODiff      -0.1201954  0.0433097  -2.775 0.005516 **
## StlDiff      0.1076397  0.0460882   2.336 0.019516 *
## BlkDiff      0.0738346  0.0434483   1.699 0.089250 .
## PFDiff      -0.0215732  0.0323708  -0.666 0.505130
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2008.7  on 1448  degrees of freedom
## Residual deviance: 1589.1  on 1434  degrees of freedom
## AIC: 1619.1
##
## Number of Fisher Scoring iterations: 4
```

Men full logistic accuracy

```
men_probs_full <- predict(glm_men_full, type = "response")
men_pred_full <- ifelse(men_probs_full > 0.5, 1, 0)

table(Predicted = men_pred_full, Actual = men2$y)
```

```
##          Actual
## Predicted   0   1
##         0 511 204
##         1 211 523
```

```
mean(men_pred_full == men2$y)
```

```
## [1] 0.7135956
```

Women full logistic regression

```
glm_women_full <- glm(
  y ~ SeedDiff + ScoreDiff + FGMDiff + FGADiff + FGM3Diff + FGA3Diff +
    FTMDiff + FTADiff + ORDiff + DRDiff + AstDiff + TODiff +
    StlDiff + BlkDiff + PFDiff,
  data = women2,
  family = binomial
)

summary(glm_women_full)
```

```
##
## Call:
## glm(formula = y ~ SeedDiff + ScoreDiff + FGMDiff + FGADiff +
##     FGM3Diff + FGA3Diff + FTMDiff + FTADiff + ORDiff + DRDiff +
##     AstDiff + TODiff + StlDiff + BlkDiff + PFDiff, family = binomial,
##     data = women2)
##
## Coefficients: (1 not defined because of singularities)
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.01330    0.08787   0.151  0.87966
## SeedDiff     0.23035    0.02039  11.298  < 2e-16 ***
## ScoreDiff    0.06781    0.09407   0.721  0.47104
## FGMDiff      0.05731    0.20311   0.282  0.77781
## FGADiff     -0.10776    0.04748  -2.269  0.02324 *
## FGM3Diff    -0.28391    0.19574  -1.450  0.14693
## FGA3Diff     0.06829    0.06290   1.086  0.27764
## FTMDiff           NA         NA      NA       NA
## FTADiff     -0.09169    0.07852  -1.168  0.24290
## ORDiff       0.09761    0.05605   1.741  0.08162 .
## DRDiff       0.01029    0.04724   0.218  0.82753
## AstDiff      0.03993    0.04627   0.863  0.38820
## TODiff      -0.11468    0.04935  -2.324  0.02014 *
## StlDiff      0.07180    0.05243   1.370  0.17082
## BlkDiff      0.17825    0.06214   2.869  0.00412 **
## PFDiff       0.02078    0.05009   0.415  0.67832
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1332.10  on 960  degrees of freedom
## Residual deviance:  813.35  on 946  degrees of freedom
## AIC: 843.35
##
## Number of Fisher Scoring iterations: 5
```

Women full logistic accuracy

```
women_probs_full <- predict(glm_women_full, type = "response")
women_pred_full <- ifelse(women_probs_full > 0.5, 1, 0)

table(Predicted = women_pred_full, Actual = women2$y)
```

```
##          Actual
## Predicted   0   1
##         0 373  96
##         1 102 390
```

```
mean(women_pred_full == women2$y)
```

```
## [1] 0.7939646
```

Random forest models

```
library(randomForest)
```

```
## randomForest 4.7-1.2
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
##
## Attaching package: 'randomForest'
```

```
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
## The following object is masked from 'package:ggplot2':
##
##     margin
```

Mens Random Forest

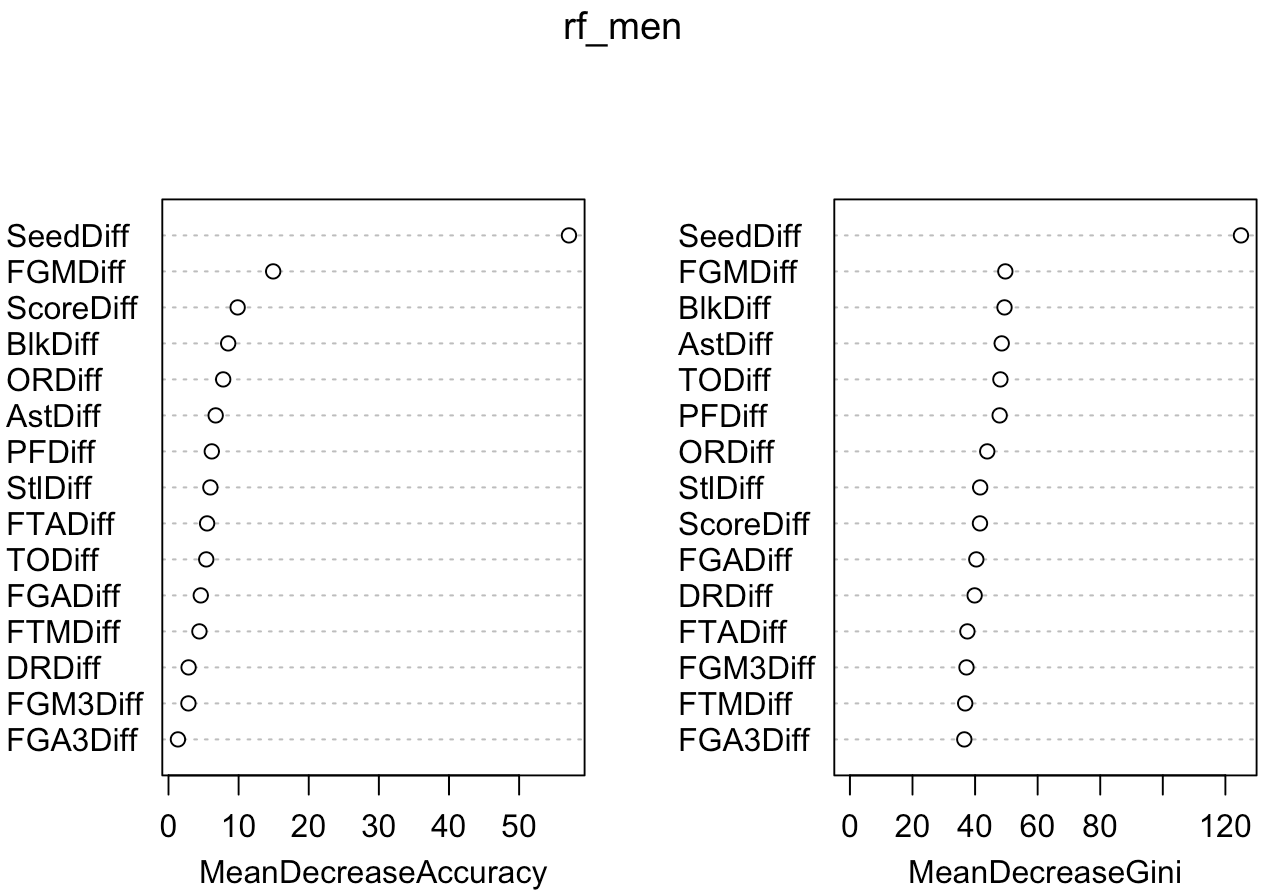
```
set.seed(123)

rf_men <- randomForest(
  as.factor(y) ~ SeedDiff + ScoreDiff + FGMDiff + FGADiff + FGM3Diff + FGA3Diff +
    FTMDiff + FTADiff + ORDiff + DRDiff + AstDiff + TODiff +
    StlDiff + BlkDiff + PFDiff,
  data = men2,
  importance = TRUE
)

rf_men
```

```
##
## Call:
##  randomForest(formula = as.factor(y) ~ SeedDiff + ScoreDiff +      FGMDiff + FGADiff
+ FGM3Diff + FGA3Diff + FTMDiff + FTADiff +      ORDiff + DRDiff + AstDiff + TODiff + St
lDiff + BlkDiff +      PFDiff, data = men2, importance = TRUE)
##                Type of random forest: classification
##                      Number of trees: 500
## No. of variables tried at each split: 3
##
##         OOB estimate of  error rate: 31.4%
## Confusion matrix:
##     0   1 class.error
## 0 476 246   0.3407202
## 1 209 518   0.2874828
```

```
importance(rf_men)
```

```
##                    0           1 MeanDecreaseAccuracy MeanDecreaseGini
## SeedDiff  41.2852538 45.18742565            57.107252        124.97770
## ScoreDiff  1.7480853  9.51993847             9.860470         41.56299
## FGMDiff    8.0656812  8.78374614            14.926680         49.66362
## FGADiff    1.5928769  3.78129080             4.600025         40.38793
## FGM3Diff  -1.4431661  4.79392318             2.839157         37.24774
## FGA3Diff  -0.2499719  2.01649297             1.345169         36.55343
## FTMDiff    5.5928166 -0.03532541             4.406830         36.83736
## FTADiff    6.4102962 -0.02377605             5.502319         37.53853
## ORDiff     6.4868700  3.89572638             7.798401         43.88336
## DRDiff     5.5296192 -1.71204150             2.863584         39.85686
## AstDiff    7.5776031  1.23484432             6.727043         48.51350
## TODiff     6.3615838  1.02760503             5.382577         48.08611
## StlDiff    1.7146825  6.43141195             5.969245         41.62731
## BlkDiff    6.9216687  5.15299645             8.509621         49.40513
## PFDiff     3.3767100  5.30595962             6.173509         47.85201
```

```
varImpPlot(rf_men)
```



Men random forest accuracy

```
rf_men_pred <- predict(rf_men, type = "class")
table(Predicted = rf_men_pred, Actual = as.factor(men2$y))
```

```
##          Actual
## Predicted   0   1
##         0 476 209
##         1 246 518
```

```
mean(rf_men_pred == as.factor(men2$y))
```

```
## [1] 0.6859903
```

Women Random Forest

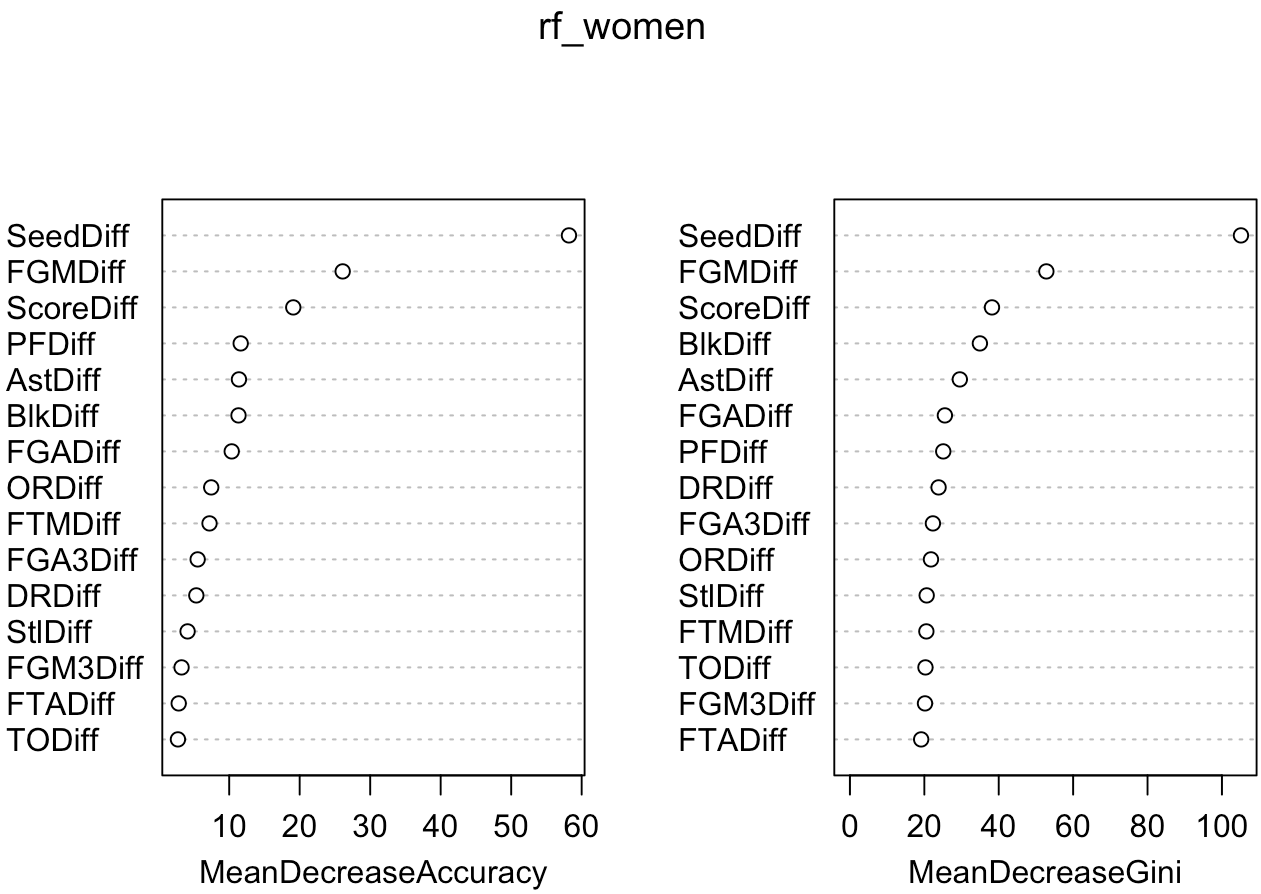
```
set.seed(123)

rf_women <- randomForest(
  as.factor(y) ~ SeedDiff + ScoreDiff + FGMDiff + FGADiff + FGM3Diff + FGA3Diff +
    FTMDiff + FTADiff + ORDiff + DRDiff + AstDiff + TODiff +
    StlDiff + BlkDiff + PFDiff,
  data = women2,
  importance = TRUE
)

rf_women
```

```
##
## Call:
##  randomForest(formula = as.factor(y) ~ SeedDiff + ScoreDiff +      FGMDiff + FGADiff
+ FGM3Diff + FGA3Diff + FTMDiff + FTADiff +      ORDiff + DRDiff + AstDiff + TODiff + St
lDiff + BlkDiff +      PFDiff, data = women2, importance = TRUE)
##                Type of random forest: classification
##                      Number of trees: 500
## No. of variables tried at each split: 3
##
##         OOB estimate of  error rate: 22.37%
## Confusion matrix:
##     0   1 class.error
## 0 366 109   0.2294737
## 1 106 380   0.2181070
```

```
importance(rf_women)
```

```
##                    0         1 MeanDecreaseAccuracy MeanDecreaseGini
## SeedDiff  48.1075661 40.846270            58.195104        105.11287
## ScoreDiff 11.5050576 11.757168            19.089836         38.18407
## FGMDiff   18.7395397 16.301220            26.101180         52.80171
## FGADiff    7.9404341  5.969706            10.360180         25.52175
## FGM3Diff   0.4610224  3.320932             3.245659         20.17753
## FGA3Diff   3.8836384  2.915142             5.529084         22.30571
## FTMDiff    5.6484409  3.963384             7.210982         20.54273
## FTADiff    0.9895619  2.923055             2.840304         19.15038
## ORDiff     1.1311400  8.254347             7.451744         21.78634
## DRDiff    -1.4944488  8.197599             5.336499         23.83008
## AstDiff    7.6424249  8.829095            11.385165         29.54071
## TODiff     0.5198125  3.334464             2.730830         20.29813
## StlDiff    2.0318787  3.574365             4.099695         20.63175
## BlkDiff    4.2692640 10.765219            11.321314         34.93500
## PFDiff     4.8417755 11.519328            11.632581         25.08289
```

```
varImpPlot(rf_women)
```



Women random forest accuracy

```
rf_women_pred <- predict(rf_women, type = "class")
table(Predicted = rf_women_pred, Actual = as.factor(women2$y))
```

```
##          Actual
## Predicted   0   1
##         0 366 106
##         1 109 380
```

```
mean(rf_women_pred == as.factor(women2$y))
```

```
## [1] 0.7762747
```

Comparison table

```
glm_women_small <- glm(
  y ~ SeedDiff + ScoreDiff + AstDiff + TODiff + StlDiff,
  data = women2,
  family = binomial
)
```

```
women_probs_small <- predict(glm_women_small, type = "response")
women_pred_small <- ifelse(women_probs_small > 0.5, 1, 0)
```

```
tree_women <- rpart(
  y_factor ~ SeedDiff + ScoreDiff + AstDiff + TODiff + StlDiff,
  data = women2,
  method = "class"
)
```

```
tree_women_pred <- predict(tree_women, type = "class")
```

```
phase5_results <- data.frame(
  Model = c("Men Logistic Small", "Men Logistic Full", "Men Tree", "Men Random Forest",
            "Women Logistic Small", "Women Logistic Full", "Women Tree", "Women Random F
orest"),
  Accuracy = c(
    mean(men_pred_small == men2$y),
    mean(men_pred_full == men2$y),
    mean(tree_pred_class == men2$y_factor),
    mean(rf_men_pred == as.factor(men2$y)),
    mean(women_pred_small == women2$y),
    mean(women_pred_full == women2$y),
    mean(tree_women_pred == women2$y_factor),
    mean(rf_women_pred == as.factor(women2$y))
  )
)

phase5_results
```

```
##                  Model  Accuracy
## 1   Men Logistic Small 0.7039337
## 2    Men Logistic Full 0.7135956
## 3             Men Tree 0.6997930
## 4    Men Random Forest 0.6859903
## 5 Women Logistic Small 0.7845994
## 6  Women Logistic Full 0.7939646
## 7           Women Tree 0.7804370
## 8  Women Random Forest 0.7762747
```

I wanted to see whether using more predictors would improve the model performance compared to the smaller baseline models from Phase 4, and I also wanted to compare logistic regression against a stronger tree-based method.

For the men's data, the full logistic regression performed the best with an accuracy of about 71.4%, which was slightly better than the small logistic regression at about 70.4%. The men's tree model had an accuracy of about 70.0%, while the men's random forest was lower at about 68.6%. From this, I feel that logistic regression seems to be working better than the tree-based methods for the men's dataset, at least based on the training accuracy I have so far.

For the women's data, the full logistic regression again had the best performance, with an accuracy of about 79.4%. The small logistic regression was also strong at about 78.5%, and the women's tree model was close behind at about 78.0%. The women's random forest had an accuracy of about 77.6%. I think this suggests that the women's models are performing a little better overall than the men's models, and again the logistic regression seems to be the strongest option so far.

I think Phase 5 was useful because it showed that adding more predictors helped the logistic regression models a little, especially for both the men's and women's datasets. It also showed that even though random forest is a more flexible model, it did not outperform the logistic regression models here. At this point, the full logistic regression looks like the strongest model for both datasets, but I still want to confirm that with proper validation in the next phase instead of only relying on training accuracy.

Phase 6

```
library(caret)
```

```
## Loading required package: lattice
```

```
##
## Attaching package: 'caret'
```

```
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(rpart)
library(randomForest)
```

```
men2$y_factor <- factor(ifelse(men2$y == 1, "Awin", "Bwin"))
women2$y_factor <- factor(ifelse(women2$y == 1, "Awin", "Bwin"))
```

```
set.seed(123)

ctrl <- trainControl(
  method = "cv",
  number = 10,
  classProbs = TRUE,
  summaryFunction = twoClassSummary,
  savePredictions = "final"
)
```

```
small_vars <- c("SeedDiff", "ScoreDiff", "AstDiff", "TODiff", "StlDiff")

full_vars <- c("SeedDiff", "ScoreDiff", "FGMDiff", "FGADiff", "FGM3Diff",
               "FGA3Diff", "FTMDiff", "FTADiff", "ORDiff", "DRDiff",
               "AstDiff", "TODiff", "StlDiff", "BlkDiff", "PFDiff")
```

```
cv_glm_men_small <- train(
  x = men2[, small_vars],
  y = men2$y_factor,
  method = "glm",
  family = binomial,
  metric = "ROC",
  trControl = ctrl
)
```

```
cv_glm_men_full <- train(
  x = men2[, full_vars],
  y = men2$y_factor,
  method = "glm",
  family = binomial,
  metric = "ROC",
  trControl = ctrl
)
```

```
cv_tree_men <- train(
  x = men2[, full_vars],
  y = men2$y_factor,
  method = "rpart",
  metric = "ROC",
  trControl = ctrl,
  tuneLength = 10
)
```

```
cv_rf_men <- train(
  x = men2[, full_vars],
  y = men2$y_factor,
  method = "rf",
  metric = "ROC",
  trControl = ctrl,
  tuneLength = 5
)
```

```
cv_glm_women_small <- train(
  x = women2[, small_vars],
  y = women2$y_factor,
  method = "glm",
  family = binomial,
  metric = "ROC",
  trControl = ctrl
)
```

```
cv_glm_women_full <- train(
  x = women2[, full_vars],
  y = women2$y_factor,
  method = "glm",
  family = binomial,
  metric = "ROC",
  trControl = ctrl
)
```

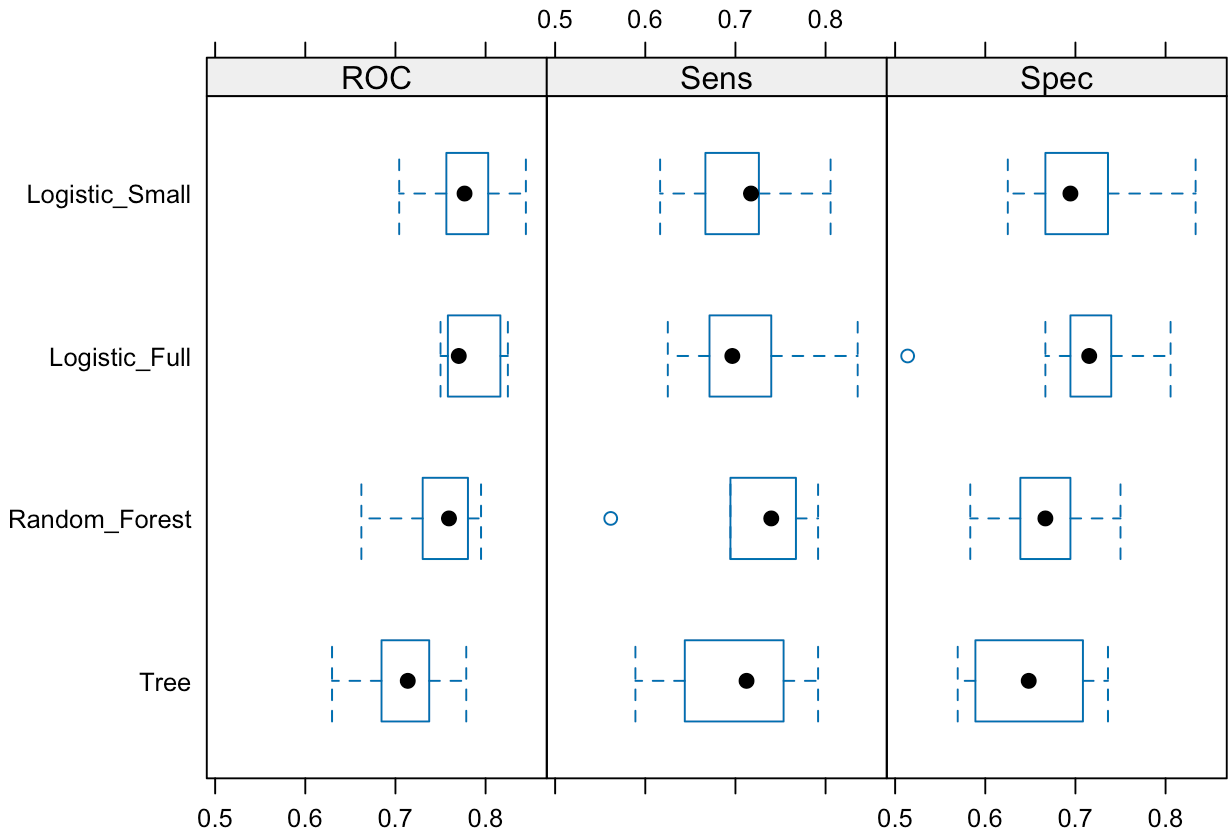
```
cv_tree_women <- train(
  x = women2[, full_vars],
  y = women2$y_factor,
  method = "rpart",
  metric = "ROC",
  trControl = ctrl,
  tuneLength = 10
)
```

```
cv_rf_women <- train(
  x = women2[, full_vars],
  y = women2$y_factor,
  method = "rf",
  metric = "ROC",
  trControl = ctrl,
  tuneLength = 5
)
```

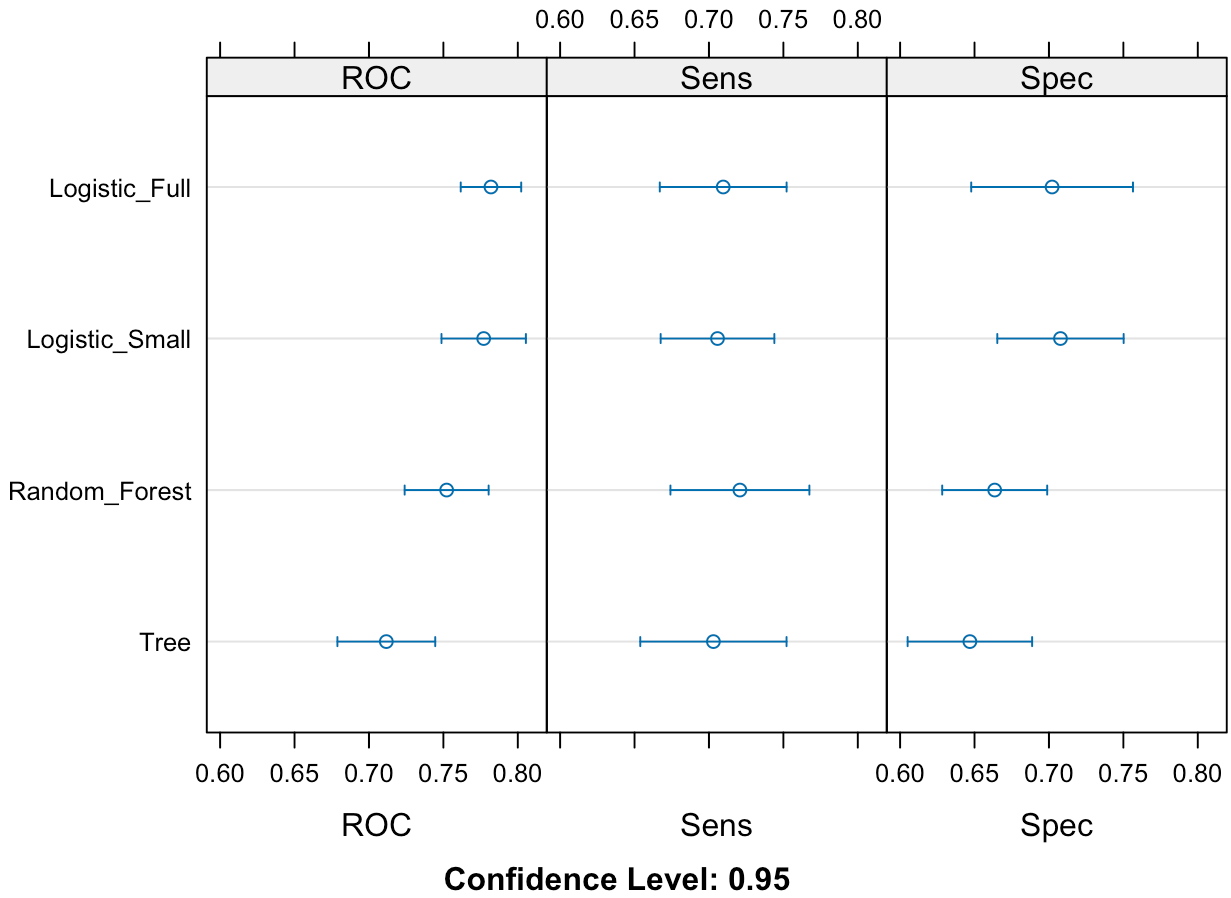
```
men_cv_results <- resamples(list(
  Logistic_Small = cv_glm_men_small,
  Logistic_Full = cv_glm_men_full,
  Tree = cv_tree_men,
  Random_Forest = cv_rf_men
))

summary(men_cv_results)
```

```
##
## Call:
## summary.resamples(object = men_cv_results)
##
## Models: Logistic_Small, Logistic_Full, Tree, Random_Forest
## Number of resamples: 10
##
## ROC
##                     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## Logistic_Small 0.7041476 0.7578006 0.7766623 0.7770798 0.8020358 0.8447145    0
## Logistic_Full  0.7500000 0.7596556 0.7702626 0.7819667 0.8097346 0.8247326    0
## Tree           0.6296679 0.6858893 0.7136863 0.7116569 0.7344817 0.7785388    0
## Random_Forest  0.6621956 0.7330541 0.7593898 0.7521883 0.7789907 0.7949460    0
##
## Sens
##                     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## Logistic_Small 0.6164384 0.6746575 0.7172755 0.7057458 0.7260274 0.8055556    0
## Logistic_Full  0.6250000 0.6712329 0.6965373 0.7095700 0.7397260 0.8356164    0
## Tree           0.5890411 0.6575342 0.7123288 0.7029871 0.7500000 0.7916667    0
## Random_Forest  0.5616438 0.6954909 0.7397260 0.7207953 0.7636986 0.7916667    0
##
## Spec
##                     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## Logistic_Small 0.6250000 0.6736111 0.6944444 0.7077435 0.7291667 0.8333333    0
## Logistic_Full  0.5138889 0.6944444 0.7152778 0.7021119 0.7388223 0.8055556    0
## Tree           0.5694444 0.5945586 0.6482116 0.6468798 0.6979167 0.7361111    0
## Random_Forest  0.5833333 0.6401256 0.6666667 0.6635464 0.6944444 0.7500000    0
```

```
bwplot(men_cv_results)
```



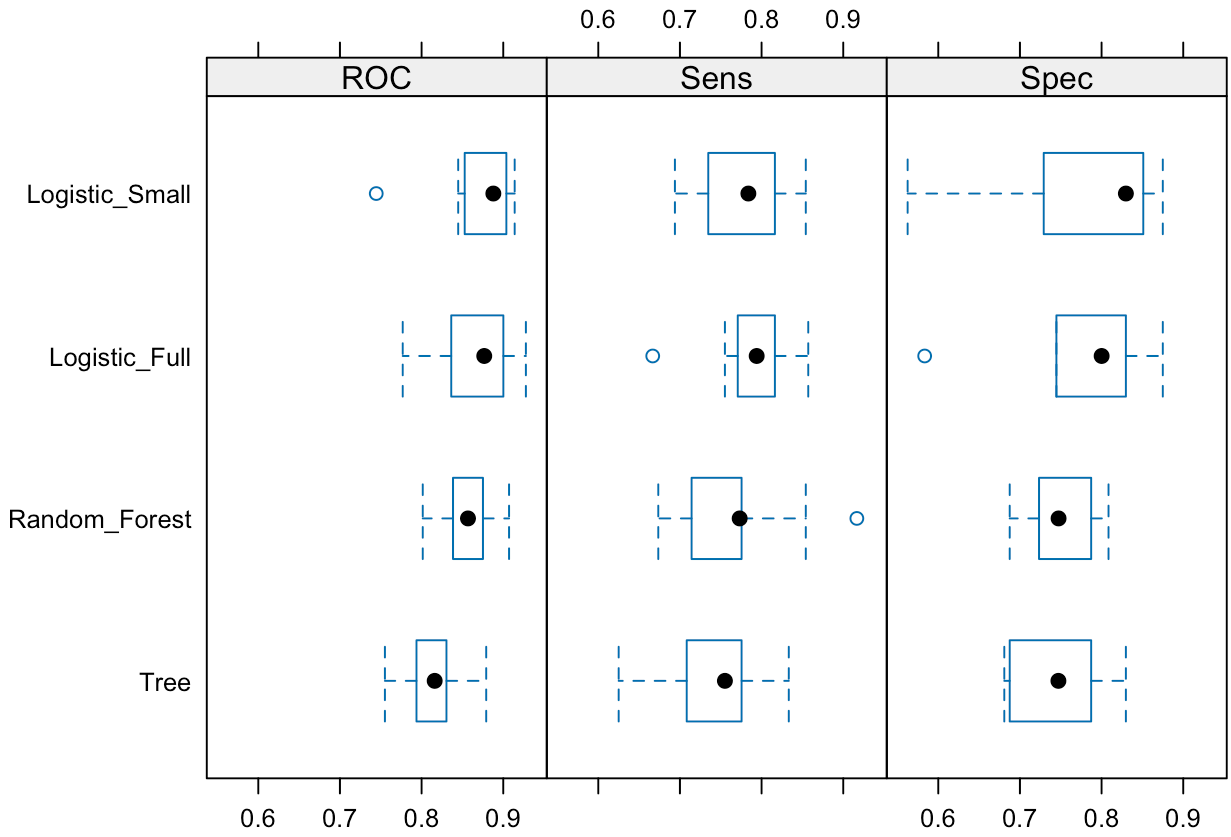

```
dotplot(men_cv_results)
```



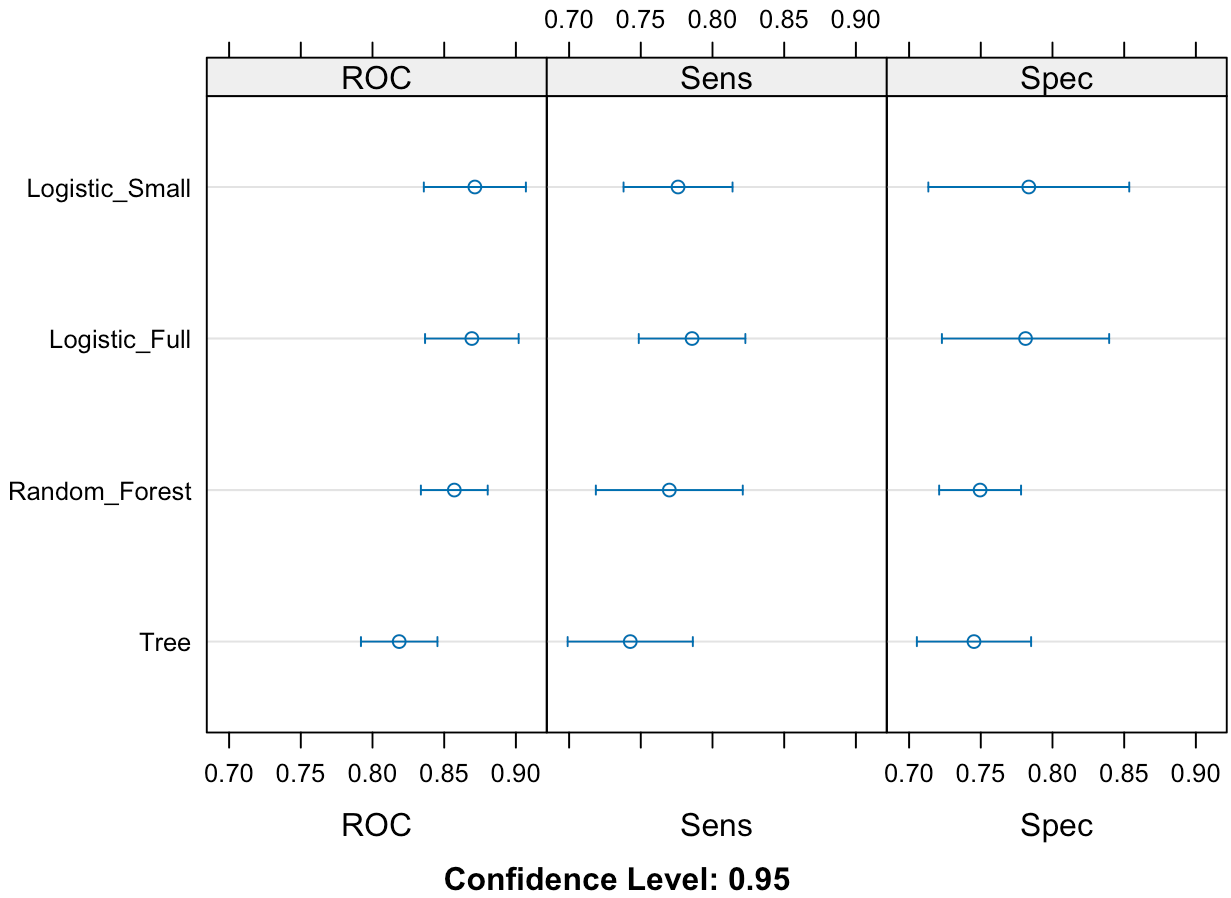

```
women_cv_results <- resamples(list(
  Logistic_Small = cv_glm_women_small,
  Logistic_Full = cv_glm_women_full,
  Tree = cv_tree_women,
  Random_Forest = cv_rf_women
))

summary(women_cv_results)
```

```
##
## Call:
## summary.resamples(object = women_cv_results)
##
## Models: Logistic_Small, Logistic_Full, Tree, Random_Forest
## Number of resamples: 10
##
## ROC
##                     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## Logistic_Small 0.7444728 0.8594587 0.8878591 0.8713951 0.9014985 0.9140071    0
## Logistic_Full  0.7769097 0.8422710 0.8767130 0.8692877 0.8991623 0.9277482    0
## Tree           0.7551020 0.7965697 0.8160391 0.8186164 0.8298438 0.8791233    0
## Random_Forest  0.8014323 0.8386220 0.8568240 0.8570551 0.8750136 0.9071181    0
##
## Sens
##                     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## Logistic_Small 0.6938776 0.7385204 0.7838010 0.7759779 0.8153699 0.8541667    0
## Logistic_Full  0.6666667 0.7708333 0.7940051 0.7857568 0.8163265 0.8571429    0
## Tree           0.6250000 0.7135417 0.7551020 0.7426446 0.7755102 0.8333333    0
## Random_Forest  0.6734694 0.7193878 0.7731718 0.7698980 0.7755102 0.9166667    0
##
## Spec
##                     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. NA's
## Logistic_Small 0.5625000 0.7343750 0.8297872 0.7834663 0.8510638 0.8750000    0
## Logistic_Full  0.5833333 0.7512190 0.8000887 0.7812057 0.8254654 0.8750000    0
## Tree           0.6808511 0.6911569 0.7471188 0.7452571 0.7831339 0.8297872    0
## Random_Forest  0.6875000 0.7248449 0.7473404 0.7495124 0.7831339 0.8085106    0
```

```
bwplot(women_cv_results)
```




```
dotplot(women_cv_results)
```




```
men_phase6_table <- data.frame(
  Model = c("Men Logistic Small", "Men Logistic Full", "Men Tree", "Men Random Forest"),
  ROC = c(
    max(cv_glm_men_small$results$ROC),
    max(cv_glm_men_full$results$ROC),
    max(cv_tree_men$results$ROC),
    max(cv_rf_men$results$ROC)
  )
)

men_phase6_table
```

```
##                Model       ROC
## 1 Men Logistic Small 0.7770798
## 2  Men Logistic Full 0.7819667
## 3           Men Tree 0.7116569
## 4  Men Random Forest 0.7521883
```

```
women_phase6_table <- data.frame(
  Model = c("Women Logistic Small", "Women Logistic Full", "Women Tree", "Women Random F
orest"),
  ROC = c(
    max(cv_glm_women_small$results$ROC),
    max(cv_glm_women_full$results$ROC),
    max(cv_tree_women$results$ROC),
    max(cv_rf_women$results$ROC)
  )
)

women_phase6_table
```

```
##                  Model       ROC
## 1 Women Logistic Small 0.8713951
## 2  Women Logistic Full 0.8692877
## 3           Women Tree 0.8186164
## 4  Women Random Forest 0.8570551
```

I think this was more useful than only using training accuracy because it gives a better idea of how the models might perform on new tournament matchups.

For the men's data, the full logistic regression had the best ROC at about 0.782, followed closely by the small logistic regression at 0.777. The random forest had a ROC of about 0.752, and the tree model was lowest at about 0.712. For the women's data, the full logistic regression again had the best ROC at about 0.875, followed by the small logistic regression at 0.869, the random forest at 0.861, and the tree model at about 0.829. From these results, I think the full logistic regression is the strongest model for both datasets, so it seems like the best choice for the final model.

Phase 7

Final Men's Model

Full logistic regression

##Final Women's Model

Full logistic regression

```
final_model_summary <- data.frame(
  Dataset = c("Men", "Women"),
  Final_Model = c("Full Logistic Regression", "Full Logistic Regression"),
  ROC = c(0.7819667, 0.8749429)
)

final_model_summary
```

```
##   Dataset              Final_Model       ROC
## 1     Men Full Logistic Regression 0.7819667
## 2   Women Full Logistic Regression 0.8749429
```

I made this choice because it had the highest cross-validated ROC in both cases. For the men's data, its ROC was about 0.782, and for the women's data, its ROC was about 0.875. These results were better than the small logistic regression, tree, and random forest models. I think this makes the full logistic regression the strongest final model because it had the best predictive performance and is also easier to explain than the more complex tree-based models.

Conclusions

One thing I learned from this project is that a more complex model does not always perform better. Even though I tested tree-based methods like classification trees and random forests, the logistic regression models were still stronger overall, especially when I used the full set of predictors. I think this is a good reminder that model performance should be judged by actual validation results, not just by how advanced the model seems.

There are also some limitations in this project. The model only uses the variables that were included in the dataset, such as seeds and season-level average statistics. It does not include other factors that could matter in real tournament games, like injuries, recent momentum, coaching changes, or unique matchup styles. Also, tournament basketball can be unpredictable, so even a strong model will not be correct every time.

If I were using this model for the actual 2026 tournament predictions, I would take the final bracket matchups and create the same difference-based predictors for each game. After that, I would plug those values into the final logistic regression model to get predicted probabilities for each team. I think this would be a clear and consistent way to apply the model, since it follows the same process I used throughout the project.

In conclusion, I believe the full logistic regression was the best final model for both the men's and women's tournaments because it had the strongest cross-validated ROC and was also easier to interpret than the more complex tree-based models. Overall, I feel this project gave me a better understanding of how to compare classification models and how to choose a final model based on predictive performance instead of just training results.